Fine-Grained Mention-Level Analysis of Biomedical Entity Linking Models

Baptiste PRAS[1] and Nona NADERI
Université Paris-Saclay, CNRS, Laboratoire Interdisciplinaire des Sciences du Numérique, 91400, Orsay, France
ORCiD ID: Baptiste PRAS https://orcid.org/0009-0004-5780-1287
ORCiD ID: Nona NADERI https://orcid.org/0000-0002-1272-7640

Abstract. Biomedical Entity Linking (BEL) is essential for structuring knowledge from biomedical texts, yet global evaluation metrics often obscure systematic model weaknesses. We propose a fine-grained evaluation framework that analyzes performance across interpretable mention-level characteristics, including length, lexical variation, synonymy, homonymy, and training frequency. Using the BELB benchmark, we apply this analysis to neural and rule-based systems. Our results show that performance degradation stems from mention-level difficulty, with consistent drops across characteristics that reflect limited training coverage, and expose model weaknesses beyond aggregate scores in a unified benchmark setting.

Keywords. entity linking, entity disambiguation, named entity, knowledge base.

1. Introduction

Biomedical Entity Linking (BEL) is the task of mapping mentions of biomedical concepts in text to standardized identifiers in a knowledge base like UMLS [1]. It is central to biomedical natural language processing, enabling downstream applications like information retrieval, clinical decision support, and knowledge base construction.

Recent neural BEL systems have achieved strong global performance, but evaluation practices remain limited. Most evaluations rely on aggregate metrics, such as recall@1, which can obscure specific failure cases. In particular, performance often varies greatly depending on mention properties, such as lexical ambiguity, surface form variability, or training frequency. To address these limitations, prior work has proposed more fine-grained evaluations. Agrawal et al. [2] introduced mention-level annotations (e.g., frequency, string match) for clinical normalization tasks. Varma et al. [3] examined the impact of surface similarity, ambiguity, and frequency on accuracy. More recently, the BELB benchmark [4] unified evaluation across eight BEL datasets with consistent data processing and metrics. Building on these efforts, we propose a general analysis framework that can be integrated directly into BELB. We automatically annotate test mentions with interpretable features—mention length, lexical variation, synonymy, homonymy, and training frequency—and analyze model performance across these dimensions. We apply our framework to two contrasting neural models: ArboEL [5], a dual-encoder architecture with coreference modeling via arborescence, and

[1] Corresponding Author: Baptiste PRAS, baptiste.pras@universite-paris-saclay.fr

GenBioEL [6], a generative model that produces entity IDs token by token. Our experiments uncover model-specific weaknesses, provide insights into generalization, and offer practical tools to guide model improvement. To facilitate future research on this task, we release the source code and data of our experiments.[2]

2. Methods

2.1. Characteristics, Binning, and Evaluation Protocol

We define the characteristics of mentions and bin the mentions into categories:

Mention length: number of tokens in the mention, using the BERT-base-uncased tokenizer. Mentions with ≤ 10 tokens are classified as short; otherwise, as long.

Lexical variation: whether the mention text appears exactly as one of the names of the gold concept in the corresponding knowledge base (KB). We compute the normalized Levenshtein distance between the mention and the gold concept's closest synonym. Mentions are labeled low (distance ≤ 0.1) or high lexical variation.

Synonymy: number of distinct names (synonyms) for the gold concept in the KB. Concepts with ≤ 10 are considered low synonymy; otherwise, as high synonymy.

Homonymy: number of different concepts that share a close (normalized Levenshtein distance ≤ 0.1), or exact name in the dictionary. A mention is labeled as homonymous if its surface form is associated with more than one concept in the KB.

Zero-shot / Rare name: whether the exact mention string was rare or unseen during training. E.g., "HIV" is considered rare if it appears only a few times (≤ 10 times), and zero-shot if never seen; otherwise, it is frequent.

Zero-shot / Rare entities: whether the gold concept was rare or unseen during training. E.g., the concept "C0282687" for Ebola is zero-shot if it never appears in training. Concepts seen ≤ 10 are classified as rare; otherwise, as frequent.

Each of these characteristics is computed automatically from the KB of the dataset and the training set. These characteristics reflect both the mention difficulty (i.e., form or ambiguity) and availability in the training data. We evaluate model performance across categories (e.g., short vs. long mentions, or seen vs. unseen names). This gives us a more detailed view of strengths and weaknesses compared to a global score.

2.2. Biomedical Entity Linking Models

Among BELB's neural approaches, we focus our analysis on two top-performing representative architectures: GenBioEL and ArboEL. *ArboEL* is a discriminative model based on a BERT encoder and is trained to score correct mention-mention and mention-entity edges above incorrect ones, enforcing global consistency. *GenBioEL* is a generative approach formulating entity linking as a sequence generation task. It uses a BART encoder-decoder architecture and is trained to generate the surface form of the correct entity token given a sentence containing the mention. To efficiently constrain generation, GenBioEL restricts decoding to valid entity names by organizing candidate surface forms into a prefix-based structure, allowing only valid completions during

[2] https://github.com/baptistepras/BEL-Mention-Analysis

generation. In addition, we include a rule-based model (RBES) to provide a non-neural reference. Each RBES is tailored to its entity type and relies on lexical rules and string-matching heuristics to link mentions to concept names in the corresponding KB.

2.3. Datasets and Evaluation Procedure

We evaluate the models on three biomedical entity linking datasets: Linnaeus [7], S800 [8], and MedMentions [9]. These datasets were selected for their diversity in domain coverage. All models are evaluated using BELB, which ensures consistency in evaluation by unifying the format of datasets, providing a common KB, and standardizing the evaluation protocol across all datasets and models. The evaluation metric used is recall@1, which measures the proportion of instances where the top ranked candidate exactly matches the gold-standard entity.

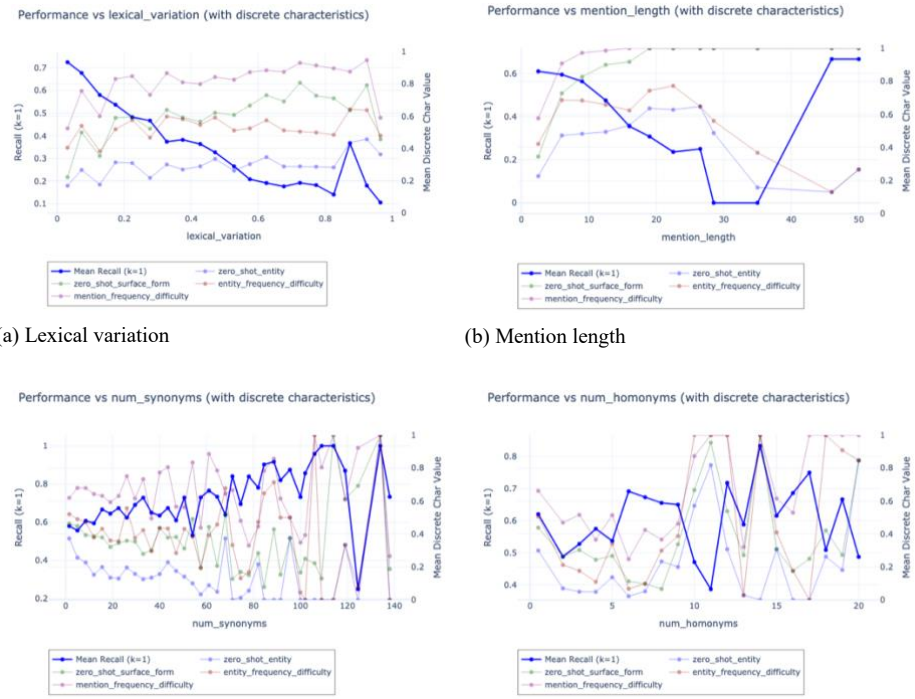
3. Results

Table 1 presents the performance of ArboEL, GenBioEL, and rule-based (RBES) models across the mention characteristics introduced in Section 2.1. For each difficult case (e.g., high lexical variation), we report the model performance together with the drop relative to the corresponding easy condition (e.g., low lexical variation). To assess whether these degradations are statistically reliable, we estimate 95% confidence intervals for the performance drop via bootstrap resampling over mentions, allowing us to determine whether the observed drops are unlikely to be due to random variation.

Table 1. Performance over mention characteristics "difficult case". Drops relative to the corresponding "easy" cases are shown in red. For each model, 95% bootstrap confidence intervals are reported for the drop.

Characteristic	GenBioEL	IC95%	ArboEL	IC95%	RBES	IC95%
Zero-shot entity	$0.54_{\downarrow 0.13}$	[0.12, 0.14]	$0.48_{\downarrow 0.27}$	[0.26, 0.28]	$\mathbf{0.56}_{\downarrow 0.05}$	[0.03, 0.06]
Zero-shot name	$\mathbf{0.55}_{\downarrow 0.16}$	[0.15, 0.17]	$0.51_{\downarrow 0.31}$	[0.30, 0.32]	$0.48_{\downarrow 0.19}$	[0.18, 0.20]
Long mention	$\mathbf{0.55}_{\downarrow 0.09}$	[0.05, 0.14]	$0.47_{\downarrow 0.25}$	[0.20, 0.29]	$0.40_{\downarrow 0.19}$	[0.15, 0.24]
Few synonyms	$0.57_{\downarrow 0.15}$	[0.14, 0.16]	$\mathbf{0.64}_{\downarrow 0.10}$	[0.10, 0.11]	$0.53_{\downarrow 0.14}$	[0.13, 0.15]
Homonyms exist	$0.60_{\downarrow 0.07}$	[0.06, 0.08]	$\mathbf{0.69}_{\downarrow 0.01}$	[0.00, 0.02]	$0.53_{\downarrow 0.11}$	[0.10, 0.12]
High lexical variation	$0.41_{\downarrow 0.39}$	[0.38, 0.40]	$\mathbf{0.45}_{\downarrow 0.42}$	[0.41, 0.43]	$0.26_{\downarrow 0.56}$	[0.55, 0.57]
Rare entity	$\mathbf{0.60}_{\downarrow 0.08}$	[0.08, 0.09]	$\mathbf{0.60}_{\downarrow 0.17}$	[0.16, 0.17]	$0.58_{\downarrow 0.02}$	[0.01, 0.02]
Rare mention	$\mathbf{0.61}_{\downarrow 0.08}$	[0.07, 0.09]	$\mathbf{0.61}_{\downarrow 0.23}$	[0.22, 0.24]	$0.55_{\downarrow 0.12}$	[0.11, 0.13]

Across most mention characteristics, all models exhibit substantial performance drops with narrow 95% bootstrap confidence intervals, indicating systematic degradations rather than random variation. Lexical variation is the most challenging factor overall, consistently producing the largest drops across models, suggesting that diverse and non-standard surface forms are particularly difficult for both neural and RBES. Overall, these results show that mention-level difficulty factors consistently impact performance across models and should explicitly be accounted for in evaluation.



(a) Lexical variation

(b) Mention length

(c) Number of synonyms

(d) Number of homonyms

Figure 1. Performance on some characteristics, alongside the prevalence of zero-shot / rare entities and names.

4. Discussion

All models experience performance degradation under difficult conditions, but ArboEL consistently exhibits the largest drops. Its most pronounced weaknesses compared to GenBioEL and RBES occur on zero-shot and rare entities or names, where the performance gaps are substantial and the estimated drops are stable across bootstrap resampling (Table 1). These systematic degradations indicate that while ArboEL performs well when entities or surface forms are observed during training, it struggles to generalize to unseen or low-frequency cases, suggesting a tendency to overfit the training data. To further investigate this behavior, we analyze whether performance drops on other mention characteristics correlate with a higher prevalence of unseen entities or names. Figure 1a-d illustrate the relationship between model recall@1 (left y-axis) and the prevalence of zero-shot/rare cases (right y-axis) in the test set. The rightmost regions contain few mentions, making performance estimates less stable.

Across all figures, we observe a consistent relationship between performance degradation and the prevalence of unseen or rare entities or names: as model performance improves along a given characteristic, the proportion of zero-shot or rare cases decreases, and conversely increases as performance declines. This pattern mirrors the stable drops observed in Table 1 and suggests that, while some characteristics introduce additional difficulty, limited training coverage is a dominant factor underlying poor performance.

ArboEL is particularly affected by entities and names that are insufficiently represented during training. Although GenBioEL exhibits smaller drops, its gains are mainly concentrated on unseen entities and surface forms, while it remains more sensitive to intrinsic mention difficulty, such as low synonymy, homonymy, or high lexical variation. A McNemar test, averaged across characteristics and comparing ArboEL to GenBioEL performances, confirms that the performance gap between the two models is highly significant ($p = 3.55 \times 10^{-96}$), indicating that it is not due to random variation. Overall, these results suggest that mention characteristics often act as proxies for training sparsity, reinforcing the need for evaluation protocols that explicitly account for such factors.

5. Conclusion

This paper introduces a fine-grained evaluation framework for Biomedical Entity Linking that moves beyond global metrics to expose systematic model weaknesses. Although GenBioEL and ArboEL achieve strong overall performance, our experiments show that both models struggle on specific mention characteristics, particularly lexical variation and unseen or rare entities and names. We further demonstrate that performance degradation consistently correlates with limited training coverage, revealing recurring failure patterns that are hidden by aggregate scores. By extending BELB with interpretable mention-level annotations, our framework provides a practical and reproducible diagnostic tool to better understand model behavior and guide the development of more robust BEL systems.

Acknowledgement: This work has been supported by CHIST-ERA grant CHIST-ERA-22-ORD-02 and by the Agence Nationale de la Recherche under project numbers ANR-23-CHRO-0008-01 and ANR-22-CPJ1-0087-01.

References

[1] Bodenreider O. The unified medical language system (UMLS): integrating biomedical terminology. Nucleic acids research. 2004;32(suppl 1):D267-70.

[2] Agrawal M, O'Connell C, Fatemi Y, Levy A, Sontag D. Robust benchmarking for machine learning of clinical entity extraction. In: Machine Learning for Healthcare Conference. PMLR; 2020. p. 928-49.

[3] Varma M, Orr L, Wu S, Leszczynski M, Ling X, Ré C. Cross-Domain Data Integration for Named Entity Disambiguation in Biomedical Text. In: Findings of the Association for Computational Linguistics: EMNLP 2021; 2021. p. 4566-75.

[4] Garda S, Weber-Genzel L, Martin R, Leser U. BELB: a biomedical entity linking benchmark. Bioinformatics. 2023 11;39(11):btad698.

[5] Agarwal D, Angell R, Monath N, Mccallum A. Entity Linking via Explicit Mention-Mention Coreference Modeling. In: Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies; 2022. p. 4644-58.

[6] Yuan H, Yuan Z, Yu S. Generative Biomedical Entity Linking via Knowledge Base-Guided Pre-training and Synonyms-Aware Fine-tuning. In: Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies; 2022. p. 4038-48.

[7] Gerner M, Nenadic G, Bergman CM. LINNAEUS: a species name identification system for biomedical literature. BMC bioinformatics. 2010;11(1):85.

[8] Pafilis E, Frankild SP, Fanini L, Faulwetter S, et al. The SPECIES and ORGANISMS resources for fast and accurate identification of taxonomic names in text. PloS one. 2013;8(6):e65390.

[9] Mohan S, Li D. MedMentions: A large biomedical corpus annotated with umls concepts. arXiv preprint arXiv:1902.09476. 2019.